

Crime Trends in West Yorkshire: A Data Visualisation Report

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Abstract

This report uses datasets from the UK Police Data Portal to analyse crime patterns in West Yorkshire for the year 2025. This dataset has a total of 300,547 observations and 12 variables. The dataset contains monthly records of reported crimes which include the outcome, location, crime type and month. In this report crime type, month and location will be used as variables to investigate trends in criminal activity throughout West Yorkshire.

The data visualisation techniques which are applied will identify the most frequent types of crimes, how the levels change over time and which locations throughout West Yorkshire experienced higher rates in crime.

1. Introduction

Crime is something that happens everyday whether we want it to or not and it can impact people and communities a lot whether it's directly or indirectly.

The crime rate in West Yorkshire is estimated to be 98 crimes per 1,000 people with violence and sexual offences being the most common having a report rate of 46 reports per 1000 with a crime index 1.5 times higher than the national average. (CrimeRate.co.uk 2026)

The aim of this report is to present an overview of crime distribution using visualisation methods to find key trends and patterns within the data set. These outcomes can help to clarify the trends and patterns between crime behaviour and crime distribution within West Yorkshire.

1.1 Dataset Description

This data set was obtained from the UK Police Data Portal which is a public access site that provides crime data in England but for the purposes of this analysis I used data specifically from West Yorkshire Police for the year 2025. (UK Police Data Portal, 2025)

This is a reliable data set because it comes straight from the UK Police database which is an official government platform. It contained around 300,000 observations which were condensed to focus on the three key variables which are month, crime type and location. (UK Police Data Portal, 2025)

It's important to highlight that only reported crimes are included in this dataset which can change the results found in this analysis as it doesn't reflect all criminal activity throughout West Yorkshire.

2. Data Processing

2.1 Combining .csv files and creating the data frame

The data set was originally split into multiple files which represented crime data for each month throughout 2025. To resolve this so that the data was easy to interpret the .csv files were merged using the **rbind()** function which allowed for the data to be one combined dataset. This was a crucial step to complete as it allowed for me to complete an effective analysis for the full year instead of separate months. The code used to perform this is listed in appendix B.

2.2 Variable selection and data reduction

In the original dataset there were 12 variables which weren't all appropriate to the focus of this analysis. To improve the clarity and efficiency of the dataset, I reduced it down to the three key variables which were Month, Crime type, and Location. I used the base R subsetting **[]** operator along with the **c()** function which selected the specific columns that were needed to focus on the relevant information for this analysis which allowed it to be presented clearly. The code that was used to perform this is listed in appendix C.

2.3 Handling of missing data

A lot of the rows in the location column of the dataset were recorded as "No Location" which I decided to remove from the data using the base R subsetting in combination with a logical condition (**!=**). This allowed for my analysis to only focus on the reports that did include location descriptions. It's worth mentioning that although these reports still had information about the crime type and month, I decided it was best to exclude them to keep consistency across the dataset. This can lead to a small decrease in total numbers of the dataset, and this is demonstrated in appendix D.

2.4 Grouping and transforming variables

I grouped several of the crime categories into wider categories to improve the simplicity and understanding of the dataset when creating the plots. When I completed a test plot it was obvious that some categories had lower amounts of crime compared to others or had similar descriptions which made it less effective and messier when presented separately in my plots.

I combined categories in the crime type column such as other crime and other theft into one named other. I also combined the less repeated offences robbery, theft from the person and bicycle theft into a wider category called minor theft offences. This allowed for these categories to still be included in the plots and make sure that the

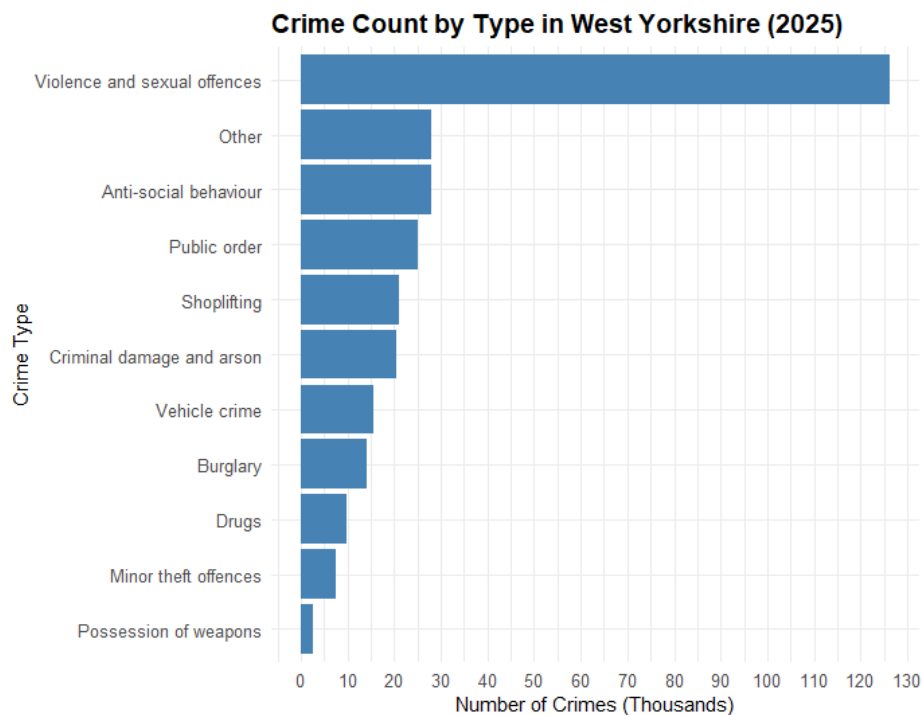
correlation between crime types was clearly illustrated. To do this I used the **mutate()** and **case_when()** function which combined the categories into the new wider ranged ones and this code is shown in full in Appendix E.

3. Research Questions

I prepared this report around a list of research questions which will explore crime patterns in West Yorkshire during 2025. I used plotting techniques to detect key trends and provide a clear understanding of the dataset. Below is the list of questions which have been addressed:

- What were the most common crimes in West Yorkshire within that year?
- Which locations in West Yorkshire recorded the highest number of crimes?
- How did crime levels change across the months of 2025?

3.1 Most common crimes



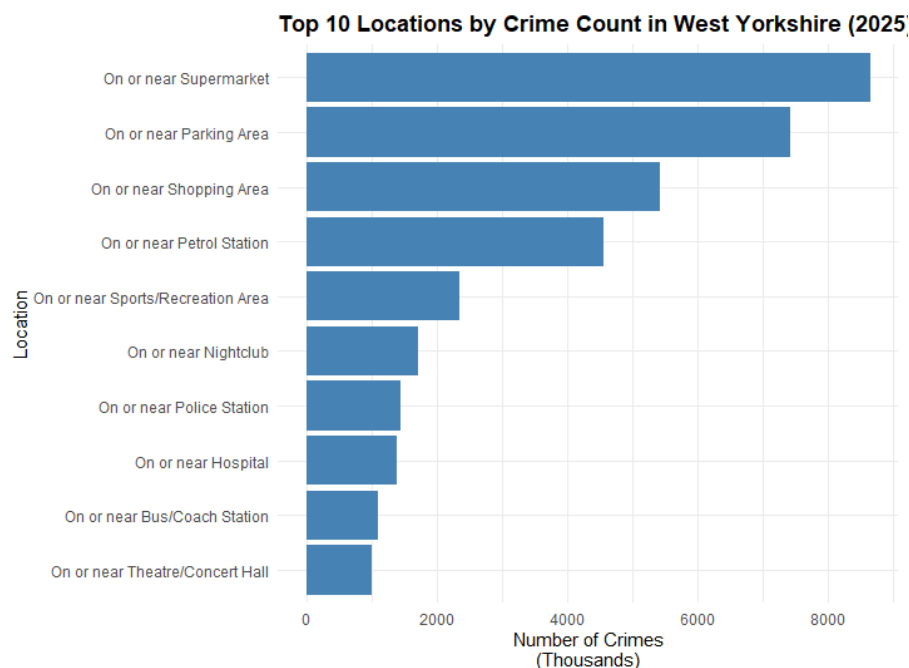
This plot shows a clear indicator that violence and sexual offences were the top crimes during 2025 with “other” and “anti-social behaviour” categories following behind.

To find out the most common crime in West Yorkshire I first started by grouping all the information in the crime type column using the **count()** function which showed the total number of reported crimes for each category, further demonstration of this is shown in Appendix F. I used **ggplot()** to create a bar chart which I decided to present horizontally instead of vertically due to the category names being longer in length, this was done by using the **coord_flip()** function. This makes the plot easier to read and understand for others.

In general, the most common crime was violence and sexual offences. This can give the police an insight into which crimes happen more frequently which would allow them to concentrate their efforts on prevention and resource allocation.

```
> ggplot(crime_counts_grouped, aes(x = reorder('Crime type', n), y = n / 1000)) +  
+   geom_bar(stat = "identity", fill = "steelblue") +  
+   coord_flip() +  
+   scale_y_continuous(breaks = seq(0, 130, by = 10)) +  
+   ggtitle("Crime Count by Type in West Yorkshire (2025)") +  
+   labs(  
+     x = "Crime Type",  
+     y = "Number of Crimes (Thousands)"  
+   ) +  
+   theme_minimal() +  
+   theme(  
+     plot.title = element_text(size = 14, face = "bold"),  
+     axis.title = element_text(size = 11),  
+     axis.text = element_text(size = 10)  
+   )
```

3.2 Top 10 crime locations



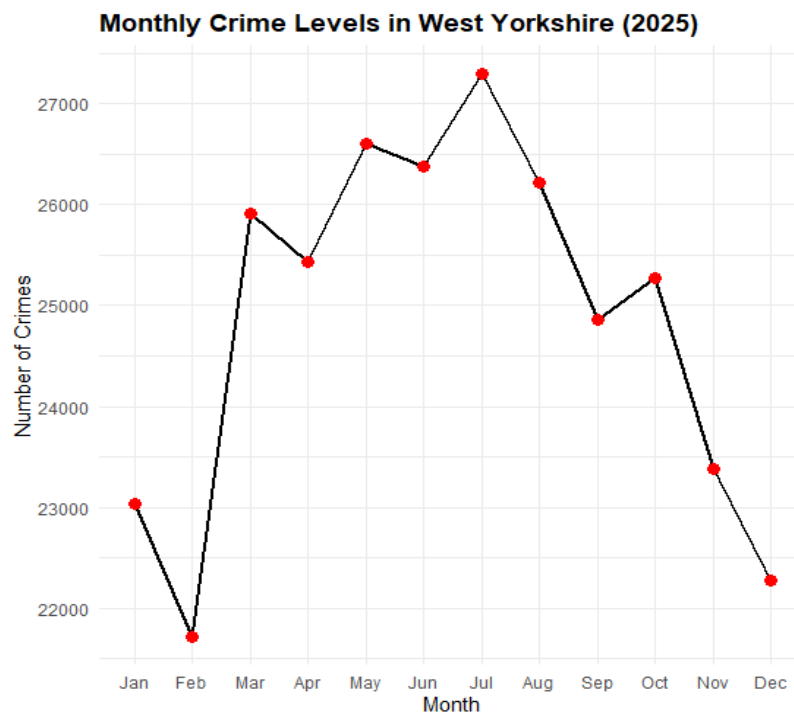
This plot represents the top 10 areas throughout West Yorkshire where crime occurs. I used the same techniques used previously to create this bar plot and then identified the top 10 by using previous functions to count and sort the data. This is shown in appendix H.

This shows that crime levels are higher around areas with a lot of human activity such as retail and commercial locations. This could be because opportunity is higher for offences like theft and anti-social behaviour to happen in busy public places where there are always interactions happening between people.

Lastly, crimes which stated “No Location” descriptions were not included in my analysis as it wouldn’t have provided meaningful information regarding crime locations.

```
> ggplot(location_counts, aes(x = reorder(Location, n), y = n)) +  
+   geom_bar(stat = "identity", fill = "steelblue") +  
+   coord_flip() +  
+   scale_y_continuous(breaks = seq(0, 9000, by = 2000)) +  
+   ggtitle("Top 10 Locations by Crime Count in West Yorkshire (2025)") +  
+   labs(  
+     x = "Location",  
+     y = "Number of Crimes\n(Thousands)"  
+   ) +  
+   theme_minimal() +  
+   theme(  
+     plot.title = element_text(face = "bold")  
+   )
```

3.3 Crime levels across months



This plot shows how crime levels changed in West Yorkshire throughout 2025.

Using **ggplot()** I created a line chart with **geom_line()** and **geom_point()** functions to show crime trends overtime and pinpoint the total crime levels for each month. I used a minimal theme so that the plot was clear and easy to interpret. This shows that crime levels are normally lower in the colder months of the year with February recording the lowest amount of crime and July the highest.

Overall, crime gradually increases from colder to warmer months with the peak of this being midyear. This suggests that there is a possible seasonal correlation. This may be influenced by the increase in outdoor activity and social interactions during the summer months which would increase the chances of these crimes happening. On the other hand, reduced movement and activity in the colder months could be a contributing factor to lower crime levels during this time.

```
> ggplot(monthly_counts, aes(x = Month, y = n, group = 1)) +  
+   geom_line(color = "black", linewidth = 1) +  
+   geom_point(color = "red", size = 3) +  
+   scale_y_continuous(breaks = seq(22000, 28000, by = 1000)) +  
+   ggtitle("Monthly Crime Levels in West Yorkshire (2025)") +  
+   labs(  
+     x = "Month",  
+     y = "Number of Crimes"  
+   ) +  
+   theme_minimal() +  
+   theme(  
+     plot.title = element_text(face = "bold")  
+   )
```


4. Critique and future work

I believe that I provided an in-depth useful analysis which explores the crime patterns across West Yorkshire, but I also acknowledge that there were a lot of limitations which could have altered the outcome of my analysis.

If the dataset included variables such as socio-economic background or population density I could've produced a better understanding of crime patterns instead of just trends. There's also the opportunity to discover the correlation between time of day the crime occurred, or which gender committed the crime.

5. Conclusion

In conclusion, each variable was explored and discussed thoroughly as to how they influence crime levels in West Yorkshire.

The most common crimes which occurred were violence and sexual offences. The area where crime took place the most was in populated areas like supermarkets or parking lots but some of the data was missing location information so they weren't included which could cause an effect on the accuracy of my findings. I also identified that more crime occurrences happen during the warmer months.

Lastly, because the dataset only has information on crimes that were reported it doesn't show the full scope of all criminal activity.

6. Appendices

Appendix A

```
> library(readr)
> library(ggplot2)
> library(dplyr)
```

Appendix B

```
> setwd(choose.dir())
> files <- list.files(pattern = "csv", recursive = TRUE, full.names = TRUE, ignore.case = TRUE)
> crime_data <- do.call(rbind, lapply(files, read_csv))
```

Appendix C

```
> crime_small <- crime_data[, c("Month", "Crime type", "Location")]
```

Appendix D

```
> crime_filtered <- crime_small[crime_small$Location != "No Location", ]
```

Appendix E

```
> crime_grouped <- crime_filtered %>%
+   mutate(`Crime type` = case_when(
+     `Crime type` %in% c("Other crime", "other theft") ~ "Other",
+     `Crime type` %in% c("Robbery", "Theft from the person", "Bicycle theft") ~ "Minor theft offences",
+     TRUE ~ `Crime type`
+   ))
```

Appendix F

```
> count(crime_grouped, `Crime type`)
# A tibble: 11 x 2
  `Crime type`      n
  <chr>          <int>
1 Anti-social behaviour 27869
2 Burglary             14131
3 Criminal damage and arson 20531
4 Drugs                9949
5 Minor theft offences   7570
6 Other               27914
7 Possession of weapons  2565
8 Public order         25042
9 Shoplifting          21006
10 Vehicle crime       15476
11 Violence and sexual offences 126332
```

Appendix G

```
> crime_grouped %>%  
+   count(Month)  
# A tibble: 12 x 2  
  Month      n  
  <chr>  <int>  
1 2025-01 23038  
2 2025-02 21716  
3 2025-03 25910  
4 2025-04 25437  
5 2025-05 26601  
6 2025-06 26378  
7 2025-07 27296  
8 2025-08 26219  
9 2025-09 24863  
10 2025-10 25272  
11 2025-11 23381  
12 2025-12 22274
```

Appendix H

```
> location_counts %>%  
+   arrange(desc(n))  
# A tibble: 10 x 2  
  Location      n  
  <chr>  <int>  
1 On or near Supermarket      8656  
2 On or near Parking Area     7426  
3 On or near Shopping Area    5428  
4 On or near Petrol Station    4574  
5 On or near Sports/Recreation Area 2349  
6 On or near Nightclub       1721  
7 On or near Police Station    1451  
8 On or near Hospital         1390  
9 On or near Bus/Coach Station  1102  
10 On or near Theatre/Concert Hall 1006
```

7. References

Crimerate.co.uk. (2026). *West Yorkshire Police Statistics*.

<https://crimerate.co.uk/west-yorkshire>

UK Police Data Portal. (2026). *Data downloads*. <https://data.police.uk/data/>